

Q1 - 24 June - Shift 1

The set of all values of k for which

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$(\tan^{-1} x)^3 + (\cot^{-1} x)^3 = k\pi^3, x \in \mathbb{R}$, is the interval :

(A) $\left[\frac{1}{32}, \frac{7}{8}\right]$

(B) $\left(\frac{1}{24}, \frac{13}{16}\right)$

(C) $\left[\frac{1}{48}, \frac{13}{16}\right]$

(D) $\left[\frac{1}{32}, \frac{9}{8}\right]$

Q2 - 24 June - Shift 2

Let $x*y = x^2 + y^3$ and $(x*1)*1 = x*(1*1)$.

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Then a value of $2 \sin^{-1} \left(\frac{x^4 + x^2 - 2}{x^4 + x^2 + 2} \right)$ is

(A) $\frac{\pi}{4}$

(B) $\frac{\pi}{3}$

(C) $\frac{\pi}{2}$

(D) $\frac{\pi}{6}$

Q3 - 25 June - Shift 2

The value of $\tan^{-1} \left(\frac{\cos\left(\frac{15\pi}{4}\right) - 1}{\sin\left(\frac{\pi}{4}\right)} \right)$ is equal to

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(A) $-\frac{\pi}{4}$

(B) $-\frac{\pi}{8}$

(C) $-\frac{5\pi}{12}$

(D) $-\frac{4\pi}{9}$

Questions

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Q4 - 26 June - Shift 2

If the inverse trigonometric functions take
principal values, then

$\cos^{-1}\left(\frac{3}{10}\cos\left(\tan^{-1}\left(\frac{4}{3}\right)\right)+\frac{2}{5}\sin\left(\tan^{-1}\left(\frac{4}{3}\right)\right)\right)$ is equal

to :

- (A) 0 (B) $\frac{\pi}{4}$ (C) $\frac{\pi}{3}$ (D) $\frac{\pi}{6}$

Space for your notes:

Q5 - 27 June - Shift 1

$\sin^{-1}\left(\sin\frac{2\pi}{3}\right)+\cos^{-1}\left(\cos\frac{7\pi}{6}\right)+\tan^{-1}\left(\tan\frac{3\pi}{4}\right)$ is

equal to :

- (A) $\frac{11\pi}{12}$ (B) $\frac{17\pi}{12}$
(C) $\frac{31\pi}{12}$ (D) $-\frac{3\pi}{4}$

Space for your notes:

Q6 - 27 June - Shift 2

The value of $\cot\left(\sum_{n=1}^{50}\tan^{-1}\left(\frac{1}{1+n+n^2}\right)\right)$ is

- (A) $\frac{26}{25}$ (B) $\frac{25}{26}$
(C) $\frac{50}{51}$ (D) $\frac{52}{51}$

Space for your notes:

Q7 - 29 June - Shift 1

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$$50 \tan \left(3 \tan^{-1} \left(\frac{1}{2} \right) + 2 \cos^{-1} \left(\frac{1}{\sqrt{5}} \right) \right) + 4\sqrt{2} \tan \left(\frac{1}{2} \tan^{-1} (2\sqrt{2}) \right) \text{ is equal to } \underline{\hspace{2cm}}.$$

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Answer Key

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Q1 (A)

Q2 (B)

Q3 (B)

Q4 (C)

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Q5 (A)

Q6 (A)

Q7 (29)

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Q1 (A)

$$\text{Let } S = (\tan^{-1} x)^3 + (\cot^{-1} x)^3 \\ = (\tan^{-1} x + \cot^{-1} x) - 3 \tan^{-1} x \cdot \cot^{-1} x (\tan^{-1} x + \cot^{-1} x)$$

$$= \frac{\pi^3}{8} - \frac{3\pi}{2} \tan^{-1} x \left(\frac{\pi}{2} - \tan^{-1} x \right)$$

$$= \frac{3\pi}{2} \left(\tan^{-1} x - \frac{\pi}{4} \right)^2 + \frac{\pi^3}{32}$$

$$\Rightarrow \frac{\pi^3}{32} \leq S < \frac{7}{8} \pi^3$$

$$= \frac{\pi^3}{32} \leq K \pi^3 < \frac{7}{8} \pi^3$$

$$\frac{1}{32} \leq K < \frac{7}{8}$$

Q2 (B)

$$\therefore (x * 1) * 1 = x * (1 * 1)$$

$$(x^2 + 1) * 1 = x * (2)$$

$$(x^2 + 1)^2 + 1 = x^2 + 8$$

$$x^4 + x^2 - 6 = 0 \Rightarrow (x^2 + 3)(x^2 - 2) = 0$$

$$x^2 = 2$$

$$\Rightarrow 2 \sin^{-1} \left(\frac{x^4 + x^2 - 2}{x^4 + x^2 + 2} \right) = 2 \sin^{-1} \left(\frac{1}{2} \right)$$

$$= \frac{\pi}{3}$$

Q3 (B)

$$\tan^{-1} \left[\frac{\cos \left(4\pi - \frac{\pi}{4} \right) - 1}{\sin \frac{\pi}{4}} \right] \Rightarrow \tan^{-1} \left(\frac{\cos \frac{\pi}{4} - 1}{\sin \frac{\pi}{4}} \right)$$

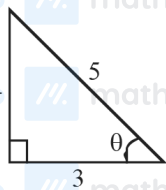
$$\tan^{-1} \left(\frac{1 - \sqrt{2}}{1} \right) = -\frac{\pi}{8}$$

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Q4 (C)

Let

$$\tan^{-1} \frac{4}{3} = \theta \Rightarrow \tan \theta = \frac{4}{3}$$



$$E = \cos^{-1} \left(\frac{3}{10} \cos \theta + \frac{2}{5} \sin \theta \right)$$

$$= \cos^{-1} \left(\frac{3}{10} \times \frac{3}{5} + \frac{2}{5} \cdot \frac{4}{5} \right)$$

$$= \cos^{-1} \left(\frac{9}{50} + \frac{8}{25} \right) = \cos^{-1} \left(\frac{25}{50} \right) = \cos^{-1} \left(\frac{1}{2} \right) = \frac{\pi}{3}$$

Q5 (A)

$$\sin^{-1} \left(\sin \frac{2\pi}{3} \right) + \cos^{-1} \left(\cos \frac{7\pi}{6} \right) + \tan^{-1} \tan \left(\frac{3\pi}{4} \right)$$

$$\sin^{-1} \sin \left(\frac{2\pi}{3} \right) = \pi - \frac{2\pi}{3} = \frac{\pi}{3}$$

$$\cos^{-1} \left(\cos \frac{2\pi}{6} \right) = 2\pi - \frac{7\pi}{6} = \frac{5\pi}{6}$$

$$\tan^{-1} \tan \left(\frac{3\pi}{4} \right) = \frac{3\pi}{4} - \pi = \frac{-\pi}{4}$$

$$\sin^{-1} \left(\sin \frac{2\pi}{3} \right) + \cos^{-1} \cos \frac{7\pi}{6} + \tan^{-1} \tan \frac{3\pi}{4}$$

$$= \frac{11\pi}{12}$$

Q6 (A)

Hints and Solutions

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$$\begin{aligned} \tan^{-1} \frac{1}{1+n+n^2} &= \tan^{-1} \left(\frac{(n+1)+n}{1+n(n+1)} \right) \\ &= \tan^{-1} (n+1) - \tan^{-1} n \\ \text{So, } \sum_{n=1}^{50} (\tan^{-1} (n+1) - \tan^{-1} n) \\ &= \tan^{-1} 51 - \tan^{-1} 1 \\ \cot \left(\sum_{n=1}^{50} \tan^{-1} \left(\frac{1}{1+n+n^2} \right) \right) &= \cot (\tan^{-1} 51 + \tan^{-1} 1) \\ &= \frac{1}{\tan(\tan^{-1} 51 - \tan^{-1} 1)} = \frac{1+51 \times 1}{51-1} = \frac{52}{50} = \frac{26}{25} \end{aligned}$$

Q7 (29)

$$\begin{aligned} &50 \tan \left(3 \tan^{-1} \frac{1}{2} + 2 \cos^{-1} \frac{1}{\sqrt{5}} \right) \\ &+ 4\sqrt{2} \tan \left(\frac{1}{2} \tan^{-1} 2\sqrt{2} \right) \\ &= 50 \tan \left(\tan^{-1} \frac{1}{2} + 2 \left(\tan^{-1} \frac{1}{2} + \tan^{-1} 2 \right) \right) \\ &+ 4\sqrt{2} \tan \left(\frac{1}{2} \tan^{-1} 2\sqrt{2} \right) \\ &= 50 \tan \left(\tan^{-1} \frac{1}{2} + 2 \cdot \frac{\pi}{2} \right) + 4\sqrt{2} \times \frac{1}{\sqrt{2}} \\ &= 50 \left(\tan \tan^{-1} \frac{1}{2} \right) + 4 \\ &= 25 + 4 = 29 \end{aligned}$$

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